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Unsupervised coefficient learning framework for variational pansharpening

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ABSTRACT

Pansharpening combines a panchromatic (PAN) image and a low-resolution multispectral (LRMS) image to generate a high-resolution multispectral (HRMS) image. Variational optimization (VO) approaches have garnered significant attention due to their data-independent generalization capabilities and robust performance. However, these methods often face challenges in accurately estimating coefficients, a critical factor influencing the quality of the final results. Existing VO approaches typically perform linear coefficient estimation at a reduced-resolution scale, which limits their effectiveness and adaptability. To address these limitations, we propose a novel VO-based method under an unsupervised coefficient learning (UCL) framework. This approach retains the generalization ability of VO while enabling precise coefficient estimation through a nonlinear, full-resolution learning technique. Furthermore, the UCL framework eliminates the need for additional training data beyond the input pair (i.e., a PAN image and a LRMS image), offering a flexible and extensible solution applicable to other traditional methods based on coefficient estimation. Qualitative and quantitative experimental assessments on reduced- and full-resolution datasets demonstrate that the proposed method achieves state-of-the-art performance. The code is available at https://github.com/Jin-liangXiao/UCL.

1. Introduction

Remote sensing images play a crucial role in various applications [1,2], including land cover monitoring [3], ground object classification [4], and anomaly detection [5]. However, due to hardware limitations, there exists a trade-off between spatial and spectral information. To address this latter, satellite systems typically acquire two types of images: low-resolution multispectral (LRMS) images, which emphasize spectral information, and high-resolution panchromatic (PAN) images, which capture fine spatial details. Panchromatic sharpening (pansharpening) is the process of fusing these image pairs to generate high-resolution multispectral (HRMS) images that combine both spectral richness and spatial accuracy.

Several methods have been proposed to address the pansharpening problem. These approaches are mainly divided into four classes [6–11]: *i*) Component substitution (CS); *ii*) Multi-resolution analysis (MRA); *iii*) Deep learning (DL); *iv*) Variational optimization (VO).

While CS and MRA methods achieve high computational efficiency, their performance is often limited. Specifically, CS methods focus on separating the spatial component from the LRMS image and substituting it with the PAN image [12–17]. Although CS methods generally preserve spatial information, they often introduce spectral distortions. In contrast, MRA methods excel at preserving spectral information while often producing spatial distortion [18–22].

Recently, DL-based methods have drawn attention due to their powerful feature extraction capabilities. PNN [23] achieved better results than traditional methods with only a three-layer network. Subsequently, a large number of DL methods, including those based on convolutional neural networks [24–28], transformers [29], and generative models [30–32], have been proposed and achieve state-of-the-art results.

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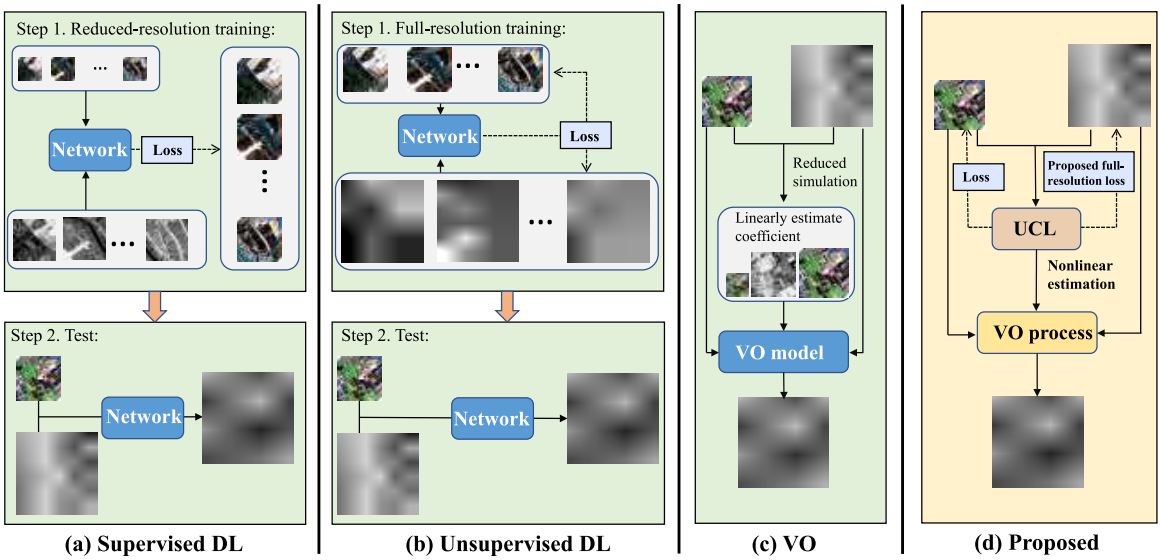

Fig. 1. The fusion processes of the supervised, unsupervised, VO, and proposed method are shown. It is clear that VO approaches, as well as the proposed method, do not require pre-training of a network. Furthermore, the proposed method can capture spatial information in a nonlinear way through the proposed UCL framework.

Fig. 2. The coefficient estimation comparison of previous methods (i.e., BT [12], GLP-HPM [52], and SFNLR [46]) and the proposed method. The PSNR values of these results are shown at the bottom. The reference $\mathcal{G}$ is computed by $\mathcal{X} \oslash \mathcal{P}$ using reduced-resolution data from the GaoFen-2 dataset. It is clear that the proposed method can achieve a better estimation of $\mathcal{G}$.

However, these methods, such as [33–35], often require extensive training data, conventionally following Wald's protocol. Their performance can be significantly impacted by a lack of sufficient training samples, a concern that is particularly relevant in remote sensing, where data scarcity is a common challenge [36–38]. Additionally, the distribution between training data and test samples plays a crucial role in the success of DL techniques. When test samples differ significantly from training data, performance can degrade sharply, which weakens the generalization ability of DL techniques [6,39] (see Section 5.4.3).

VO approaches demonstrate data-independent generalization capabilities and robust performance in the pansharpening task. Unlike DL techniques, which typically require time-consuming training, VO methods only need a single pair of inputs: a PAN image and a low-resolution multispectral (LRMS) image (see Fig. 1(c)). This allows VO methods to adapt to the unique characteristics of each test image. These methods perform fusion by constructing variational models that capture the relationship between the HRMS image and the two input images (i.e., LRMS and PAN) [7]. To model this relationship, coefficient estimation has been widely employed [40–46]. Lu et al. [47] proposed a unified model based on band-adaptive gradients and detail correction. Xiao et al. [48] introduced a context-aware detail fidelity approach with adaptive coefficient estimation. However, since the HRMS image is unknown during inference, the relationship between the estimated coefficients and the input images remains unclear, making precise estimation challenging. Many VO methods attempt to address this issue by developing diverse strategies for coefficient estimation. For example, Ballester et al. [49] assumed that the PAN image is a weighted sum of the HRMS bands, with fixed coefficients. Tian et al. [50] leveraged cartoon-texture decomposition. Fu et al. [51] used guided filtering to compute coefficients within a variational framework. Despite these efforts, most existing coefficient estimation methods are linear and operate at reduced-resolution. Due to the complexity of the underlying coefficients and the scale discrepancy between reduced and full resolution, these estimations are often inaccurate, thus limiting the fusion performance.

Bearing these concerns in mind, we propose a novel variational pansharpening method based on an unsupervised coefficient learning (UCL) framework. The UCL framework enables nonlinear coefficient estimation by introducing a new full-resolution loss function. As shown in Fig. 2, the proposed framework significantly enhances the accuracy of coefficient estimation compared to previous methods. The approach consists of two main steps. First, we accurately estimate the coefficient in the VO model using a convolutional neural network (CNN) with the proposed full-resolution loss. Second, an iterative algorithm based on the alternating direction method of multipliers (ADMM) [53] is employed to solve the variational model. Unlike supervised DL techniques, the UCL framework is trained using only a single input pair, i.e., a PAN image and a LRMS image. Thus, the proposed method is data-independent, which leads to superior generalization performance. Besides, the proposed method exploits the VO process, which inherits the good generalization ability of VO models. Extensive experimental results demonstrate that our approach achieves state-of-the-art performance.

The contributions of this paper are summarized as follows:

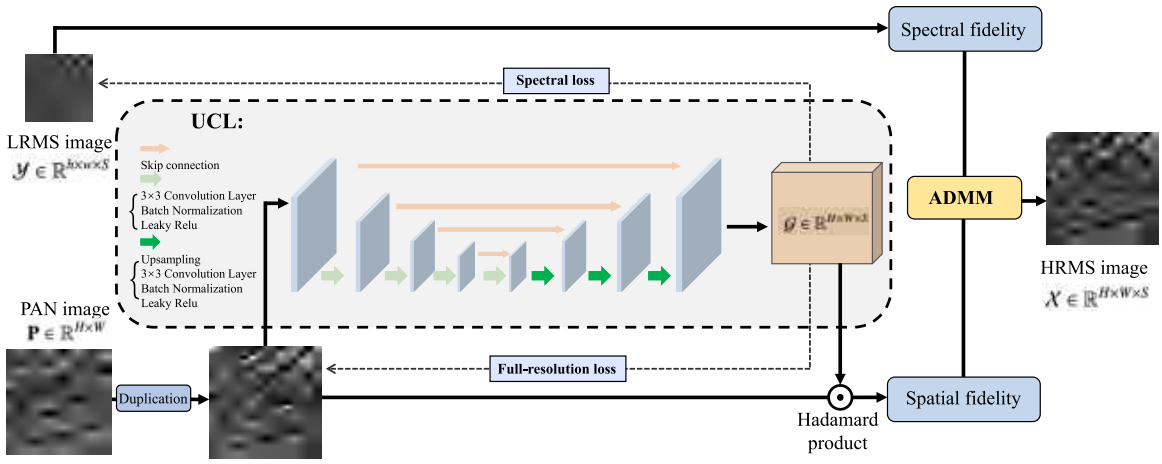

Fig. 3. Flowchart of the proposed method that firstly estimates the coefficient $\mathcal{G}$ and then injects it into the ADMM-based VO iterative process. Spectral loss and full-resolution loss are in (5) and (10), respectively. More details can be found in Section 3.

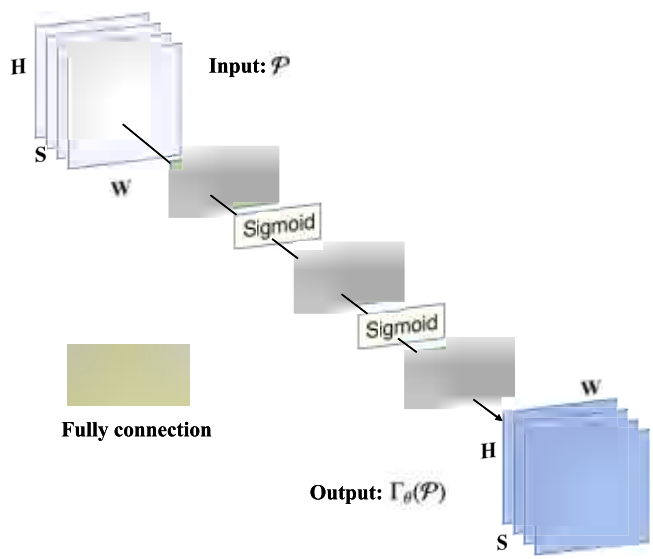

Fig. 4. The network structure of the map Γ_θ in (8).

- We propose a novel variational method based on the UCL framework, which can combine the strengths of DL and VO approaches, i.e., the nonlinear feature-extraction ability of DL techniques and the data-independent generalization of VO methods.
- The UCL framework accurately estimates the coefficient involved in spatial fidelity only using one single pair of inputs and constrains the coefficient by the proposed full-resolution loss. In addition, this framework provides a general solution for spatial fidelity of VO approaches, which can be extended to other pansharpening VO schemes.
- Experimental results on both reduced- and full-resolution datasets show the effectiveness of the proposed method compared with recent state-of-the-art pansharpening approaches.

The remainder of this paper is organized as follows. Section 2 analyzes the related works. Section 3 presents the proposed model. Section 4 describes the algorithm to effectively solve the proposed model. Section 5 shows the experiments and the related discussion. Finally, Section 6 draws concluding remarks.

2. Related works

Before describing the related works, notations, and their explanations used in this paper are reported in Table 1.

Table 1
Notations and their explanations.

Notations	Explanations
$\mathcal{A}, \mathbf{A}, \mathbf{a}, a$	Tensor, matrix, vector, scalar
$\mathbf{A}_{(3)}$	The mode-3 unfolding of $\mathcal{A}$
$\mathcal{X} \in \mathbb{R}^{H\times W\times S}$	HRMS image
$\mathcal{Y} \in \mathbb{R}^{h\times w\times S}$	LRMS image
$\widetilde{\mathcal{Y}} \in \mathbb{R}^{h\times w\times S}$	Upsampled LRMS image
$\mathbf{P} \in \mathbb{R}^{H\times W}$	PAN image
$\mathcal{P} \in \mathbb{R}^{H\times W\times S}$	Replicated PAN image
$\mathcal{P}_L \in \mathbb{R}^{H\times W\times S}$	Low-pass version of the replicated PAN image
$\mathcal{G} \in \mathbb{R}^{H\times W\times S}$	Coefficient
$\nabla_i, i = 1, 2$	The gradient operation along the i-th direction
$\nabla = \begin{bmatrix} \nabla_1 \\ \nabla_2 \end{bmatrix}$	The gradient operation.
$\odot$	Hadamard product
$\oslash$	Element-wise division

2.1. VO pansharpening methods

As analyzed in Section 1, the estimation of the coefficient for spatial fidelity is challenging and critical for the performance of VO methods. Many VO methods have been developed to compute this coefficient.

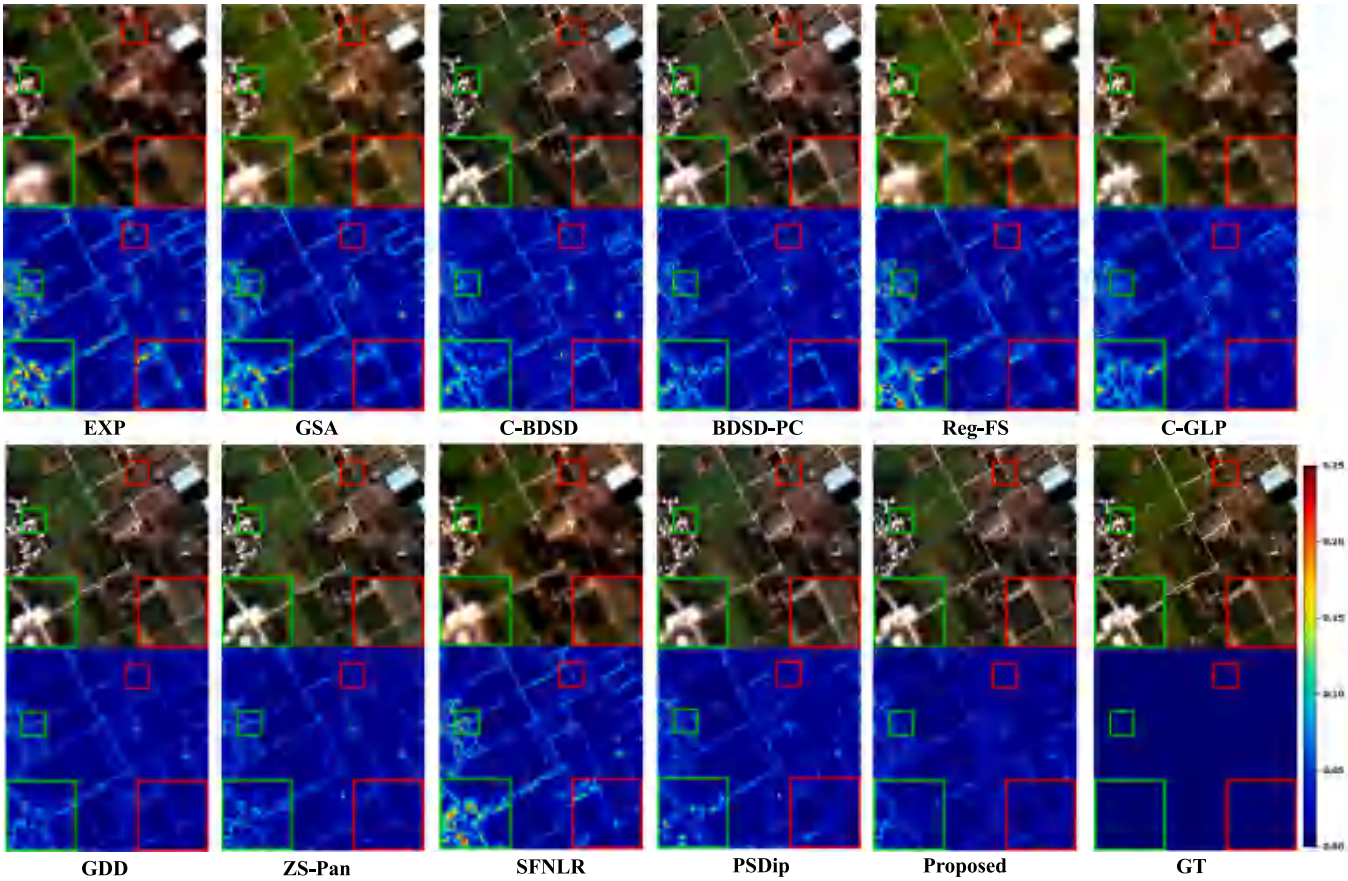

Fig. 5. The true color (3rd (R), 2nd (G), 1st (B)) and the corresponding error images of the fused products using a GaoFen-2 dataset (reduced-resolution): EXP, GSA, C-BDSD, BDSD-PC, Reg-FS, C-GLP, GDD, ZS-Pan, SFNLR, PSDip, proposed, and GT. Some close-ups are depicted in the green and red boxes.

For example, Ballester et al. [49] assumed that the PAN image is the weighted sum of each band of the HRMS image. Thus, we have:

$$\mathbf{P} = \sum_{i=1}^{S} \omega_i \mathbf{X}_i + \xi_1, \quad i = 1, \dots, S, \tag{1}$$

where $\mathbf{P} \in \mathbb{R}^{H \times W}$ is the PAN image, S represents the number of spectral bands of the HRMS image, $\mathbf{X}_i$ is the i-th band of the HRMS image $\mathcal{X}$, and ξ_1 denotes a zero-mean Gaussian noise. ω_i is the i-th value of the positive coefficient [42,54].

Fu et al. [51] proposed a spatial fidelity term with guided filter-based coefficient estimation:

$$\nabla \mathbf{x}_j = \omega_i \nabla \mathbf{p}_j + \xi_2, j \in \Omega_k, \tag{2}$$

where ∇ is the gradient operation, Ω_k represents an image block centered at location k, $\mathbf{x}_j$ and $\mathbf{p}_j$ are the HRMS and PAN block images at position j, respectively, and ξ_2 is a zero-mean Gaussian noise. ω_i is estimated by the guided filter.

A flexible spatial fidelity with a pixel-wise coefficient can be represented as follows:

$$\mathcal{X} = \mathcal{G} \odot \mathcal{P} + \xi_3, \tag{3}$$

where $\mathcal{G} \in \mathbb{R}^{H \times W \times S}$ is the coefficient, $\odot$ means Hadamard product, $\mathcal{P} \in \mathbb{R}^{H \times W \times S}$ is the replicated PAN image, and ξ_3 is a zero-mean Gaussian noise. BT [12] and GLP-HPM [52] compute the coefficient by $\widetilde{\mathcal{Y}} \oslash \mathcal{I}_L$ and $\widetilde{\mathcal{Y}} \oslash \mathcal{P}_L$, respectively, where $\mathcal{P}_L$ is the low-pass version of $\mathcal{P}$ and $\mathcal{I}_L$ is the weighted average of each band of $\widetilde{\mathcal{Y}}$. Recently, Xiao et al. [46] estimated $\mathcal{G}$ using a nonlocal regression. These methods estimate $\mathcal{G}$ exploiting the "invariance among scales" assumption, which states that the coefficients remain consistent between the reduced-resolution and full-resolution settings. However, in practice, there exists an inherent gap between the two scales. The coefficients estimated from the downsampled images may fail to capture the fine-grained spatial structures and high-frequency details that only appear at the original resolution, leading to suboptimal reconstruction performance when transferred back to the full-resolution setting.

2.2. DL pansharpening methods

DL pansharpening methods have achieved the best performance, which mainly include supervised DL approaches and unsupervised DL techniques. Supervised DL methods, e.g., [55–58], have demonstrated state-of-the-art results. Zhang et al. [27] proposed a deep variational network for blind pansharpening, which demonstrates that the variational theory of traditional methods is helpful for neural networks. Ke et al. [28] presented a kernel-guided injection deep network to handle unknown degradations. These supervised DL techniques usually first train the neural network on reduced-resolution training data and then apply the trained network to test examples. The whole process is depicted in Fig. 1(a).

Unsupervised DL methods, e.g., [32,59], train the neural network on full-resolution data, thus avoiding the exploitation of the "invariance among scale" assumption and the problems caused by the absence of ground-truth. Ma et al. [30] proposed an unsupervised framework for pansharpening based on a generative adversarial network, which does not rely on any ground-truth. He et al. [60] proposed an unsupervised pansharpening algorithm based on double-cycle consistency that can learn directly from the original MS images without relying on artificially simulated degradation processes. However, these unsupervised DL methods also heavily depend on time-consuming training process and additional training datasets, as shown in Fig. 1(b). Besides, the

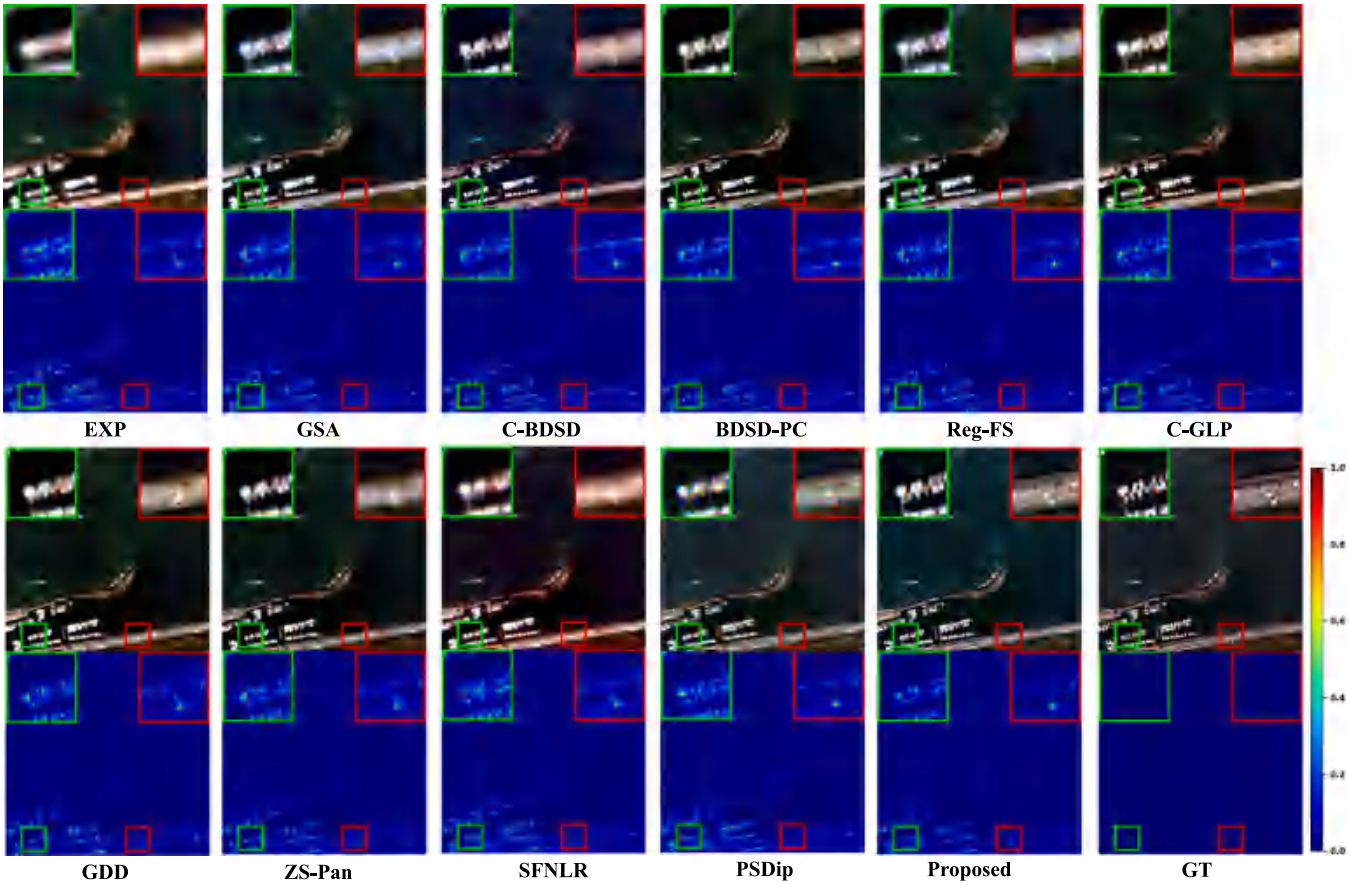

Fig. 6. The true color (3rd (R), 2nd (G), 1st (B)) and the corresponding error images of the fused products using a QuickBird dataset (reduced-resolution): EXP, GSA, C-BDSD, BDSD-PC, Reg-FS, C-GLP, GDD, ZS-Pan, SFNLR, PSDip, proposed, and GT. Some close-ups are depicted in the green and red boxes.

generalization problem of the model still exists. To overcome it, zero-shot methods have been proposed. They do not rely on additional training data. Uezato et al. [61] proposed a guided deep decoder to obtain the result using deep image prior. Cao et al. [62] proposed a zero-shot method using a training dataset generated from a single test sample. However, the relationship between the inputs and the HRMS image has still not been accurately described.

3. Proposed method

The spatial fidelity is critical for VO methods. The pixel-level framework in (3) is effective, while often leads to limited results due to the inaccuracy of the estimated coefficient (i.e., $\mathcal{G}$ in (3)). In this section, we propose a novel variational model based on the one-pair deep coefficient learning, which can combine the advantages of VO and DL methods, i.e., data independence and nonlinear feature-extraction ability.

3.1. Spectral fidelity

The LRMS image is the spatial degraded version of the HRMS image, which means that the LRMS image contains most of the spectral information of the fused product. Specifically, the LRMS image is usually considered as the HRMS image after the blurring and downsampling operation in VO-based methods [51]. Mathematically, we have:

$$\mathbf{X}_{(3)}\mathbf{BS} = \mathbf{Y}_{(3)} + \xi_4, \tag{4}$$

where $\mathbf{B} \in \mathbb{R}^{HW\times HW}$ and $\mathbf{S} \in \mathbb{R}^{HW\times hw}$ are the blurring matrix and the decimation operation, respectively, $\mathbf{Y}_{(3)} \in \mathbb{R}^{S\times hw}$ is the mode-3 unfolding [63] of $\mathcal{Y} \in \mathbb{R}^{h\times w\times S}$, which means $\mathbf{Y}_{(3)}(k,(j-1)h+i) = \mathcal{Y}(i,j,k)$, and ξ_4 indicates a zero-mean Gaussian noise. Thus, the spectral fidelity term is represented as follows:

$$f_{spec} = \|\Phi(\mathcal{X}) - \mathcal{Y}\|_F^2, \tag{5}$$

where Φ denotes the spatial degraded operation, i.e., the blurring and decimation operations **BS** in (4), and $\|\cdot\|_F$ is the Frobenius norm.

3.2. Spatial fidelity for the proposed UCL framework

The PAN image contains the spatial information of the fused product. A powerful framework to build the pixel-level relationship between the PAN and HRMS images is in (3). Thus, the spatial fidelity term is as follows:

$$f_{spa} = \|\mathcal{X} - \mathcal{G} \odot \mathcal{P}\|_F^2. \tag{6}$$

The pixel-level spatial fidelity (6) is flexible and shows high performance in the extraction of spatial features from the PAN image [46,64]. However, a significant obstacle is related to the estimation of coefficient $\mathcal{G}$. Compared with directly computing the target HRMS image, the coefficients that describe this relationship are relatively easier to estimate and have been successfully exploited in many pansharpening methods [46,51]. As discussed in Section 2.1, the coefficient in the spatial fidelity term is critical for VO methods. Because of the flexibility and a large scale of coefficient $\mathcal{G}$, this latter is always hard to be estimated in advance. Previous methods usually estimated it in a linear way exploiting the "invariance among scales" assumption [46]. Given the gap between the reduced-resolution and full-resolution scales, the estimation at reduced-resolution only describes a degraded relationship neglecting some information (as shown in Fig. 2). Hence, previous approaches

Table 2
Quantitative results for 20 reduced-resolution images using GaoFen-2 (GF-2) data. (Bold: best; Underline: second best).

Method	PSNR(± std)	SSIM (± std)	SAM (± std)	SCC (± std)	ERGAS (± std)	Q4 (± std)	Time(s)
EXP [68]	32.205 ± 2.084	0.824 ± 0.054	1.817 ± 0.367	0.928 ± 0.036	2.356 ± 0.501	0.812 ± 0.048	**0.003**
GSA [17]	34.523 ± 2.084	0.890 ± 0.036	1.804 ± 0.378	0.958 ± 0.021	1.793 ± 0.420	0.879 ± 0.032	0.012
C-BDSD [69]	34.230 ± 1.925	0.913 ± 0.025	1.888 ± 0.340	0.960 ± 0.018	1.823 ± 0.416	0.904 ± 0.024	0.116
BDSD-PC [70]	34.832 ± 2.051	0.901 ± 0.031	1.770 ±0.337	0.963 ± 0.018	1.723 ± 0.395	0.893 ± 0.028	0.011
Reg-FS [71]	34.926 ± 1.845	0.897 ± 0.031	1.758 ± 0.361	0.962 ± 0.019	1.695 ± 0.348	0.888 ± 0.029	0.018
C-GLP [72]	35.199 ± 1.974	0.908 ± 0.027	1.715 ± 0.328	0.962 ± 0.020	1.695 ± 0.356	0.898 ± 0.030	0.144
GDD [61]	34.017 ± 2.032	0.893 ± 0.032	1.598 ± 0.311	0.956 ± 0.021	1.895 ± 0.389	0.876 ± 0.026	205.1
ZS-Pan [62]	37.129 ± 1.700	**0.943 ± 0.019**	1.541 ± 0.309	0.982 ± 0.009	1.350 ± 0.301	**0.936 ± 0.022**	35.20
SFNLR [46]	37.621 ± 2.492	0.930 ± 0.026	1.325 ± 0.302	0.978 ± 0.012	1.288 ± 0.346	**0.936** ± 0.022	8.468
PSDip [64]	33.197 ± 8.091	0.854 ± 0.165	1.573 ± 0.310	0.959 ± 0.047	3.603 ± 5.387	0.821 ± 0.243	86.90
Proposed	**38.123 ± 1.624**	0.935 ± 0.016	**1.116** ± 0.204	**0.981** ± 0.010	**1.113** ± 0.192	0.935 ± 0.021	135.2
Ideal value	**+∞**	**1**	**0**	**1**	**0**	**1**	**0**

Table 3
Quantitative results for 20 reduced-resolution images using QuickBird (QB) data. (Bold: best; Underline: second best).

Method	PSNR (± std)	SSIM (± std)	SAM (± std)	SCC (± std)	ERGAS (± std)	Q4 (± std)	Time(s)
EXP [68]	28.414 ± 3.197	0.691 ± 0.089	8.514 ± 1.794	0.759 ± 0.036	11.96 ± 1.608	0.581 ± 0.079	**0.003**
GSA [17]	32.146 ± 2.393	0.852 ± 0.032	8.280 ± 1.813	0.900 ± 0.023	7.766 ± 0.592	0.812 ± 0.106	0.014
C-BDSD [69]	31.885 ± 2.555	0.857 ± 0.036	8.208 ± 1.862	0.893 ± 0.023	7.970 ± 0.691	0.820 ± 0.100	0.124
BDSD-PC [70]	32.220 ± 2.489	0.855 ± 0.035	8.243 ± 1.888	0.901 ± 0.020	7.686 ± 0.615	0.822 ± 0.095	0.013
Reg-FS [71]	32.021 ± 2.455	0.852 ± 0.034	8.026 ± 1.735	0.899 ± 0.022	7.836 ± 0.632	0.809 ± 0.101	0.018
C-GLP [72]	31.588 ± 2.576	0.836 ± 0.041	7.918 ± 1.630	0.885 ± 0.020	8.424 ± 0.676	0.794 ± 0.094	0.152
GDD [61]	30.040 ± 2.991	0.779 ± 0.060	8.173 ± 1.685	0.853 ± 0.019	10.036 ± 1.192	0.701 ± 0.098	206.7
ZS-Pan [62]	31.680 ± 2.409	0.848 ± 0.033	9.070 ± 1.835	0.889 ± 0.022	8.036 ± 0.637	0.809 ± 0.119	36.41
SFNLR [46]	32.959 ± 2.463	0.874 ± 0.029	7.336 ± 1.542	0.917 ± 0.018	7.067 ± 0.576	0.849 ± 0.088	10.44
PSDip [64]	33.957 ± 2.404	0.892 ± 0.026	6.758 ± 1.483	0.934 ± 0.015	6.309 ± 0.565	0.868 ± 0.102	87.99
Proposed	**35.025 ± 2.172**	**0.914 ± 0.018**	**6.286 ± 1.289**	**0.948 ± 0.016**	**5.563 ± 0.453**	**0.888 ± 0.097**	136.1
Ideal value	**+∞**	**1**	**0**	**1**	**0**	**1**	**0**

often provide inaccurate estimations, thus leading to spectral and spatial distortions.

To overcome the above-mentioned drawbacks, we propose a novel method to estimate the coefficient based on one-pair inputs. DL tools, e.g., convolutional neural networks, have proven effective in nonlinear feature extraction. However, DL methods are often dependent on large training datasets, and their performance can degrade if training data are insufficient or if there is a mismatch between the distributions of the training and test datasets. Recently, the deep image prior has demonstrated its potential for solving low-level image processing problems. Motivated by the no-train network format [65], we propose a novel method to estimate the coefficient based on one-pair inputs. As shown in Fig. 3, the replicated PAN image is the input of a convolutional neural network because of the similarity between the PAN and the coefficient exists. Hence, the convolutional neural network can effectively learn the coefficient from the replicated PAN image. Thus, we have:

$$\mathcal{G} = f_\Theta(\mathcal{P}), \tag{7}$$

where $f_\Theta : \mathbb{R}^{H\times W\times S} \to \mathbb{R}^{H\times W\times S}$ is the Skipnet with parameters Θ.

The next critical point is how to constrain the parameters Θ of the neural network to output the coefficient $\mathcal{G}$. Unlike from the estimation at reduced-resolution, we propose a relationship exploiting the full-resolution (original) versions of the inputs. Thus, we have:

$$f_\Theta(\mathcal{P}) = \Gamma_\theta(\mathcal{P}) \oslash \mathcal{P}, \tag{8}$$

where $\oslash$ denotes element-wise division, Γ_θ is the map from the replicated PAN image to the HRMS image and whose network structure can be shown in Fig. 4. In the gradient domain, we have:

$$\nabla f_\Theta(\mathcal{P}) = \nabla(\Gamma_\theta(\mathcal{P}) \oslash \mathcal{P}), \tag{9}$$

where ∇ is the gradient operator. Since the PAN image is closer to the HRMS image on the gradient domain than on the original domain, which has been widely demonstrated in previous works [48,66], the lightweight network, Γ_θ, can learn spatial information in the gradient domain in an easier way. Thus, we design the full-resolution loss as follows:

$$Loss_{full} = \|\nabla f_\Theta(\mathcal{P}) - \nabla(\Gamma_\theta(\mathcal{P}) \oslash \mathcal{P})\|_F^2, \tag{10}$$

which can effectively constrains the output of f_Θ. Besides, according to (6), we have

$$\mathcal{X} = \mathcal{G} \odot \mathcal{P} = f_\Theta(\mathcal{P}) \odot \mathcal{P}. \tag{11}$$

Thus, combining it with (4) and (10), the total loss is as follows:

$$Loss_{total} = \|\Phi(f_\Theta(\mathcal{P}) \odot \mathcal{P}) - \mathcal{Y}\|_F^2 + \alpha\ Loss_{full}, \tag{12}$$

where α is a balanced parameter. The total loss function contains two terms: the first term explicitly constrains Skipnet to remain consistent with Eq. (8), thereby ensuring that the coefficients it estimates align with the theoretical formulation. The other term focuses on stabilizing the auxiliary network in Fig. 4. This joint optimization strategy allows Skipnet to remain accurate while also benefiting from the auxiliary supervision provided by the additional constraint. Thus, the coefficient $\mathcal{G}$ can be effectively estimated by minimizing the total loss in (12).

3.3. The final model

Combining the above-mentioned terms, we get the final model that can be expressed as:

$$\min_{\mathcal{X}} \|\Phi(\mathcal{X}) - \mathcal{Y}\|_F^2 + \beta\|\mathcal{X} - \mathcal{G} \odot \mathcal{P}\|_F^2, \tag{13}$$

where β is a positive hyperparameter. The flowchart of the proposed model is depicted in Fig. 3. Although the final model is convex, the direct calculation leads to a high computational burden. Therefore, we design an algorithm based on the ADMM [53] framework to effectively solve the final model. The following section will be related to the proposed ADMM-based algorithm.

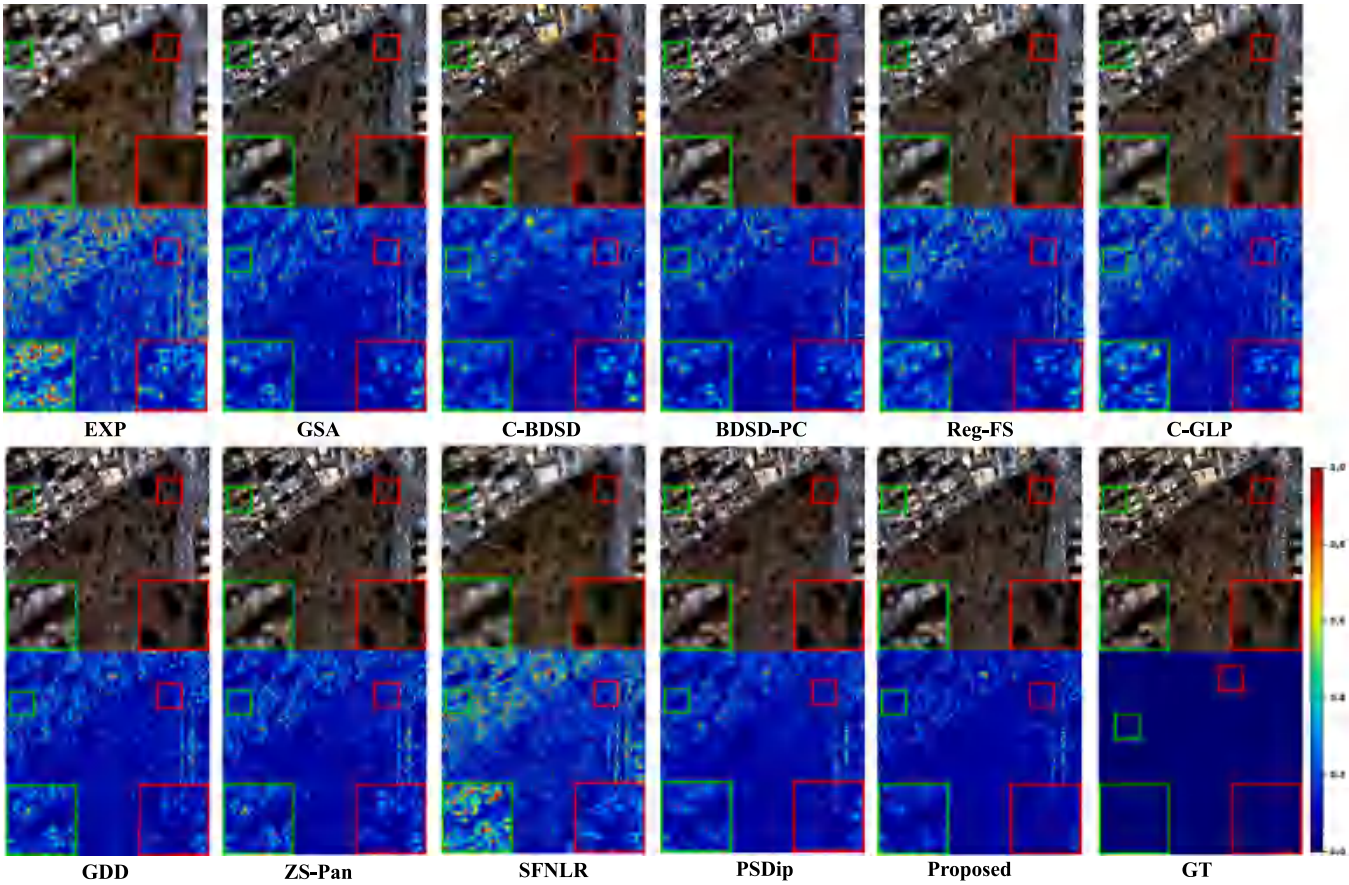

Fig. 7. The true color (5th (R), 3rd (G), 2nd (B)) and the corresponding error images of the fused products using a WorldView-3 dataset (reduced-resolution): EXP, GSA, C-BDSD, BDSD-PC, Reg-FS, C-GLP, GDD, ZS-Pan, SFNLR, PSDip, proposed, and GT. Some close-ups are depicted in the green and red boxes.

4. Model optimization

In this section, we design an effective algorithm based on the ADMM [53] framework to solve the final model in (13). We introduce the auxiliary variable $\mathbf{M} = \mathbf{X}_{(3)}\mathbf{B}$. Thus, the final model is converted into the following format:

$$\begin{aligned} &\min_{\mathbf{M},\mathcal{X}} \left\|\mathbf{MS} - \mathbf{Y}_{(3)}\right\|_F^2 + \beta\|\mathcal{X} - \mathcal{G} \odot \mathcal{P}\|_F^2 \\ &s.t. \quad \mathbf{M} = \mathbf{X}_{(3)}\mathbf{B}. \end{aligned} \tag{14}$$

From the above format, the augmented Lagrangian function is as follows:

$$\begin{aligned} \mathcal{L} = &\left\|\mathbf{MS} - \mathbf{Y}_{(3)}\right\|_F^2 + \frac{\eta}{2}\left\|\mathbf{X}_{(3)}\mathbf{B} - \mathbf{M} + \frac{\mathbf{J}_{(3)}}{\eta}\right\|_F^2 \\ &+ \beta\|\mathcal{X} - \mathcal{G} \odot \mathcal{P}\|_F^2 + const, \end{aligned} \tag{15}$$

where $\mathbf{J}_{(3)}$ is the unfolding form of the Lagrange multipliers $\mathcal{J}$, η is a positive hyperparameter, and *const* denotes a constant, which is independent of the variables. Hence, the whole model is transformed into the following three sub-problems.

4.1. M-sub-problem

According to (15), the $\mathbf{M}$ sub-problem at the $k+1$-th iteration step is expressed as the following minimization problem:

$$\min_{\mathbf{M}} \left\|\mathbf{MS} - \mathbf{Y}_{(3)}\right\|_F^2 + \frac{\eta}{2}\left\|\mathbf{X}_{(3)}^k\mathbf{B} - \mathbf{M} + \frac{\mathbf{J}_{(3)}^k}{\eta}\right\|_F^2, \tag{16}$$

where $\mathbf{X}_{(3)}^k$ denotes $\mathbf{X}_{(3)}$ at the k-th iteration step. $\mathbf{MSS}^{\mathrm{T}}$ can be represented in the following element-wise multiplication form:

$$\mathbf{MSS}^{\mathrm{T}} = \mathbf{M} \odot \mathbf{D}_{SS^{\mathrm{T}}}, \tag{17}$$

where $\cdot^{\mathrm{T}}$ denotes the transpose operator and $\mathbf{D}_{SS^{\mathrm{T}}}$ is a sparse matrix only containing zeros and ones. More details about (17) are presented in [67]. Thus, we can obtain $\mathbf{M}$ by:

$$\mathbf{M}^{k+1} = (2\mathbf{Y}_{(3)}\mathbf{S}^{\mathrm{T}} + \eta\mathbf{X}_{(3)}^k\mathbf{B} + \mathbf{J}_{(3)}^k) \oslash (2\mathbf{D}_{SS^{\mathrm{T}}} + \eta\mathbf{1}), \tag{18}$$

where $\mathbf{1}$ is an all-ones matrix.

Algorithm 1 The solution algorithm for the proposed model in (13).

Input: The LRMS image $\mathcal{Y}$, the replicated PAN image $\mathcal{P}$, the adaptive coefficient $\mathcal{G}$, the balanced parameter α, the hyperparameter β, η.

Initialization: $\mathcal{X}^0 = \Psi(\mathcal{Y}, r), \mathbf{M}^0 = \mathbf{J}_{(3)}^0 = \mathbf{0}, k = 0$

1: Update Skipnet $f_\Theta(\cdot)$ parameters Θ and Γ_θ parameters θ using the Adam optimizer by minimizing (12).
2: Compute $\mathcal{G}$ according to (7).
3: **while** $k < k_{mit} = 100$ and $RelCha > \varepsilon = 10^{-5}$ **do**
  4: Update $\mathcal{M}^{k+1}$ via (18).
  5: Update $\mathcal{X}^{k+1}$ via (20).
  6: Update Lagrange multiplier $\mathbf{J}_{(3)}^{k+1}$ via (21).
  7: $k = k + 1$.
8: **end while**

Output: The fused HRMS image $\mathcal{X}$.

Table 4
Quantitative results for 20 reduced-resolution images using WorldView-3 (WV-3) data. (Bold: best; Underline: second best).

Method	PSNR (± std)	SSIM (± std)	SAM (± std)	SCC (± std)	(± std)	Q8 (± std)	Time(s)
EXP [68]	29.144 ± 3.496	0.717 ± 0.105	5.721 ± 1.739	0.847 ± 0.059	7.059 ± 1.631	0.625 ± 0.093	**0.007**
GSA [17]	32.750 ± 2.565	0.881 ± 0.033	5.423 ± 1.682	0.932 ± 0.031	4.762 ± 1.493	0.813 ± 0.103	0.020
C-BDSD [69]	30.939 ± 3.289	0.883 ± 0.029	6.155 ± 1.621	0.892 ± 0.059	6.142 ± 2.677	0.796 ± 0.114	0.207
BDSD-PC [70]	32.897 ± 2.581	0.889 ± 0.032	5.498 ± 1.741	0.934 ± 0.029	4.685 ± 1.485	0.819 ± 0.102	0.201
Reg-FS [71]	32.430 ± 2.667	0.877 ± 0.031	5.433 ± 1.662	0.929 ± 0.030	4.903 ± 1.459	0.807 ± 0.098	0.035
C-GLP [72]	31.979 ± 2.270	0.847 ± 0.045	5.608 ± 1.802	0.915 ± 0.056	5.277 ± 1.547	0.749 ± 0.154	0.200
GDD [61]	31.044 ± 3.294	0.830 ± 0.050	5.166 ± 1.489	0.909 ± 0.037	5.633 ± 1.320	0.739 ± 0.094	206.7
ZS-Pan [62]	30.691 ± 2.191	0.841 ± 0.036	6.702 ± 1.589	0.887 ± 0.074	5.472 ± 1.189	0.743 ± 0.127	45.48
SFNLR [46]	33.482 ± 2.615	0.898 ± 0.025	4.805 ± 1.369	0.943 ± 0.027	4.354 ± 1.357	0.841 ± 0.093	17.06
PSDip [64]	34.420 ± 2.450	0.920 ± 0.019	4.439 ± 1.215	0.953 ± 0.029	3.806 ± 1.104	0.857 ± 0.095	106.1
Proposed	**35.338 ± 2.212**	**0.934 ± 0.020**	**4.151 ± 1.172**	**0.961 ± 0.027**	**3.466 ± 1.152**	**0.868 ± 0.096**	151.4
Ideal value	**+∞**	**1**	**0**	**1**	**0**	**1**	**0**

4.2. X-sub-problem

There are matrix and tensor forms of $\mathcal{X}$ in (15). For simplicity, the sub-problem in the tensor unfolding form is:

$$\min_{\mathbf{X}_{(3)}} \beta\left\|\mathbf{X}_{(3)} - \mathbf{G}_{(3)} \odot \mathbf{P}_{(3)}\right\|_F^2 + \frac{\eta}{2}\left\|\mathbf{X}_{(3)}\mathbf{B} - \mathbf{M}^{k+1} + \frac{\mathbf{J}_{(3)}^k}{\eta}\right\|_F^2. \tag{19}$$

The fast Fourier transform (FFT) algorithm is an effective method to solve the above sub-problem. Thus, we have:

$$\mathbf{X}_{(3)}^{k+1} = \mathcal{F}^{-1}\left(\mathcal{F}\left(\mathbf{H}^{k+1}\right) \oslash \mathcal{F}(\mathbf{N})\right), \tag{20}$$

with

$$\mathbf{H}^{k+1} = 2\beta\left(\mathbf{G}_{(3)} \odot \mathbf{P}_{(3)}\right) + \eta\mathbf{M}^{k+1}\mathbf{B}^{\mathrm{T}} - \mathbf{J}_{(3)}^k\mathbf{B}^{\mathrm{T}},$$

$$\mathbf{N} = 2\beta + \eta\mathbf{B}\mathbf{B}^{\mathrm{T}},$$

where $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ represent the fast Fourier transform and its inverse transformation, respectively.

4.3. Updating multipliers

Finally, the multiplier $\mathbf{J}_{(3)}$ is updated as follows:

$$\mathbf{J}_{(3)}^{k+1} = \mathbf{J}_{(3)}^k + \eta\left(\mathbf{X}_{(3)}^{k+1}\mathbf{B} - \mathbf{M}^{k+1}\right). \tag{21}$$

The relative change ($RelCha$) and the number of iterations are used as termination conditions of the algorithm. $RelCha$ is defined as:

$$RelCha = \left\|\mathbf{X}_{(3)}^{k+1} - \mathbf{X}_{(3)}^k\right\|_F / \left\|\mathbf{X}_{(3)}^k\right\|_F. \tag{22}$$

The whole solving algorithm can be summarized in Algorithm 1, where k_{mit} denotes the maximum number of iterations, r is the scale ratio between the LRMS and PAN images, ϵ refers to a tolerance value, and Ψ indicates the bicubic interpolation.

5. Experimental results

In this section, we compare the proposed method with some state-of-the-art approaches to verify the superiority of the proposed model. Qualitative and quantitative assessments are provided on PanCollection [1], which is an open train-test dataset from different sensors, i.e., GaoFen-2, QuickBird, and the WorldView series. The reduced-resolution data are obtained following Wald's protocol to test the performance of pansharpening methods. For fairness, all methods in the benchmark are run in Pytorch 2.0 and MATLAB (R2020a) on a computer of 64Gb RAM, 12th Gen Intel(R) Core(TM) i9-12900K 3.20 GHz, and NVIDIA GeForce GTX 1650. We fine-tune the parameters of all approaches to achieve the best results. Besides, more analysis about the hyperparameters, the ablation study, the comparison with advanced supervised DL methods, and the extension to other pansharpening schemes are conducted. It is worth noting that we set the number of iterations of the UCL framework to 8000, β to 10^{-3}, α to 10^{-5}, and η to 10^{-3}.

To evaluate the performance of all methods, we apply a variety of objective metrics for the comprehensive assessment. For reduced-resolution data, the peak signal-to-noise ratio (PSNR), the structural similarity index (SSIM) [73], the spectral angle mapper (SAM) [74], the spatial correlation coefficient (SCC) [75], the erreur relative globale adimensionnelle de synthèse (ERGAS) [76], the Q4 index [77] (or the Q8 [78] metric for eight-band datasets) are employed. For full-resolution data, we chose the hybrid quality with no reference (HQNR) [79], which consists of a spectral quality index, D_λ, and a spatial quality index, D_s.

5.1. Benchmark

To validate the effectiveness of the proposed model, we compare it with the following representative methods belonging to different pansharpening classes:

- CS methods:
  1. GSA: Gram-Schmidt adaptive method [17].
  2. C-BDSD: context-adaptive band-dependent spatial-detail method [69].
  3. BDSD-PC: robust band-dependent spatial-detail method [70].
- MRA methods:
  1. EXP: LRMS upsampled using a polynomial kernel with 23 coefficients [68].
  2. Reg-FS: generalized Laplacian pyramid (GLP) with a full scale regression method [71].
  3. C-GLP: the context-based GLP [72].
- DL methods:
  1. GDD: guided deep decoder [61].
  2. ZS-Pan: zero-shot semi-supervised learning for pansharpening [62].
- VO methods:
  1. SFNLR: spatial fidelity with nonlocal regression [46].
  2. PSDip: variational zero-shot method [64].

5.2. Reduced-resolution assessment

We compare the proposed method with other advanced approaches on reduced-resolution data from different sensors, i.e., GaoFen-2, QuickBird, and WorldView-3.

5.2.1. The GaoFen-2 dataset

The spatial resolution of the PAN and LRMS images captured by the GaoFen-2 sensor is 0.8 m and 3.2 m, respectively. We compare the proposed method against four different categories of pansharpening methods on 20 reduced-resolution test samples. As shown in Table 2, our method achieves the best results. We also assess the visual quality in

[1] https://github.com/liangjiandeng/PanCollection

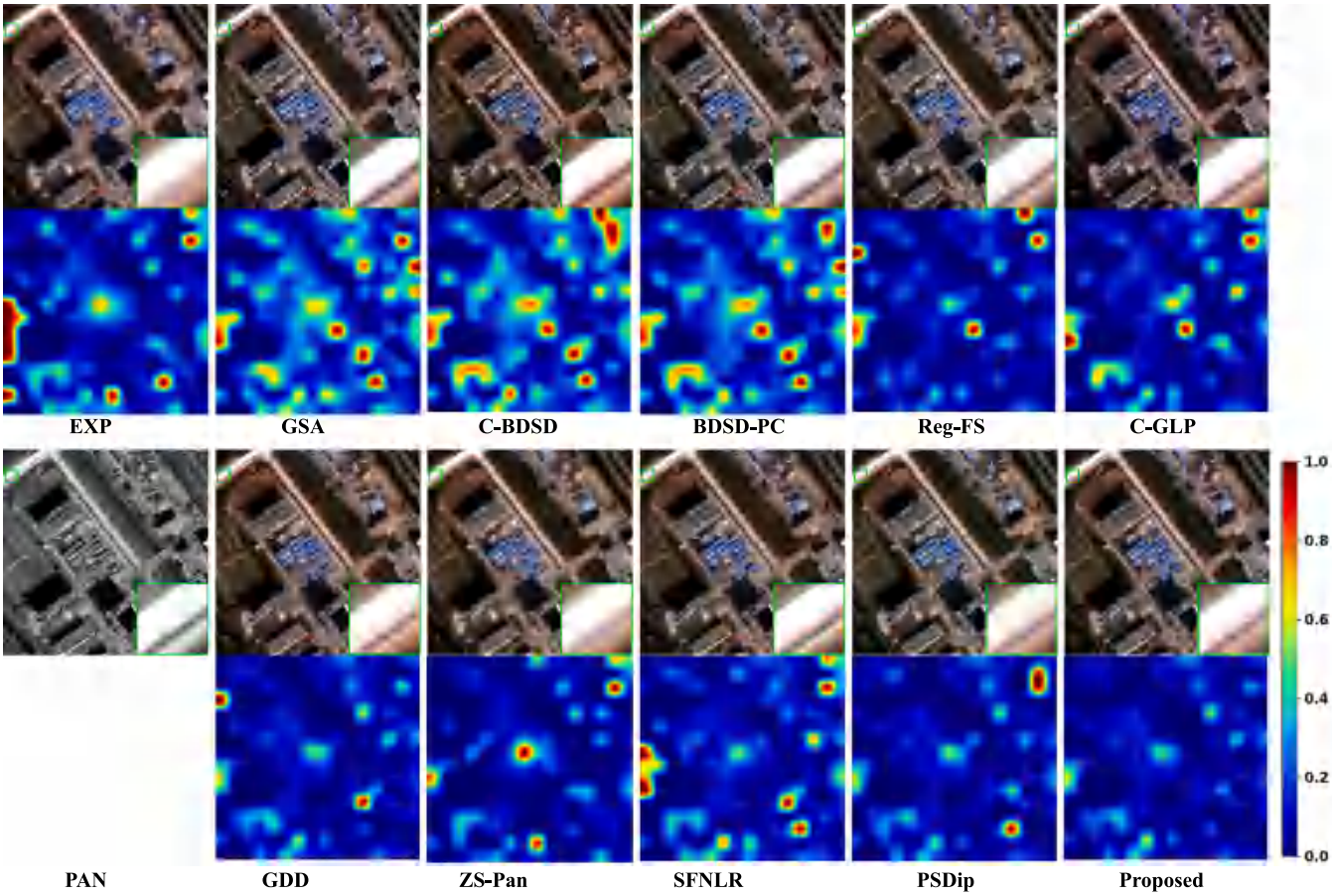

Fig. 8. Top: true color images consisting of the 3rd (R), 2nd (G), and 1st (B) bands of the fused GaoFen-2 (GF-2) dataset (full-resolution), obtained by EXP, GSA, C-BDSD, BDSD-PC, Reg-FS, C-GLP, GDD, ZS-Pan, SFNLR, PSDip, proposed, and GT. Bottom: D_λ maps of the corresponding methods.

Table 5
Quantitative results for all the compared methods in the benchmark. (a) 20 full resolution images acquired by the GaoFen-2 sensor; (b) 20 full resolution images acquired by the QuickBird sensor. (Bold: best; Underline: second best).

Method	*(a) Full GaoFen-2 Dataset*				*(b) Full QuickBird Dataset*			
	D_λ	D_s	HQNR	Time(s)	D_λ	D_s	HQNR	Time(s)
EXP [68]	0.142 ± 0.024	0.065 ± 0.023	0.802 ± 0.032	**0.016**	0.067 ± 0.033	0.124 ± 0.015	0.818 ± 0.034	**0.022**
GSA [17]	0.255 ± 0.059	0.219 ± 0.034	0.582 ± 0.052	0.076	0.083 ± 0.044	0.258 ± 0.029	0.680 ± 0.037	0.091
C-BDSD [69]	0.219 ± 0.030	0.138 ± 0.016	0.674 ± 0.033	0.640	0.097 ± 0.045	0.137 ± 0.042	0.779 ± 0.057	1.035
BDSD-PC [70]	0.202 ± 0.038	0.188 ± 0.029	0.648 ± 0.040	0.027	0.080 ± 0.042	0.197 ± 0.050	0.738 ± 0.058	0.043
Reg-FS [71]	0.118 ± 0.021	0.163 ± 0.028	0.738 ± 0.033	0.056	0.047 ± 0.020	0.175 ± 0.028	0.786 ± 0.026	0.092
C-GLP [72]	0.099 ± 0.019	0.092 ± 0.030	0.818 ± 0.033	0.738	0.070 ± 0.036	0.099 ± 0.032	0.838 ± 0.041	1.022
GDD [61]	0.172 ± 0.059	0.079 ± 0.017	0.763 ± 0.062	526.6	0.070 ± 0.034	0.133 ± 0.017	0.806 ± 0.038	531.9
ZS-Pan [62]	0.132 ± 0.029	0.098 ± 0.026	0.782 ± 0.032	33.12	0.046 ± 0.021	0.082 ± 0.022	0.876 ± 0.023	37.95
SFNLR [46]	0.100 ± 0.013	0.084 ± 0.014	0.825 ± 0.020	46.42	**0.018 ± 0.011**	0.080 ± 0.031	0.903 ± 0.030	64.18
PSDip [64]	0.078 ± 0.018	**0.032 ± 0.016**	0.892 ± 0.024	292.8	0.037 ± 0.014	0.069 ± 0.018	0.897 ± 0.026	292.9
Proposed	**0.058 ± 0.018**	0.038 ± 0.032	**0.907 ± 0.042**	560.1	0.030 ± 0.013	**0.059 ± 0.018**	**0.913 ± 0.022**	535.3
Ideal value	**0**	**0**	**1**	**0**	**0**	**0**	**1**	**0**

Table 6
The effects of the UCL framework and the VO process assessed on a QB reduced-resolution image. (Bold: best; Underline: second best).

Method	PSNR	SSIM	SAM	SCC	ERGAS	Q4
Replicated PAN image	28.620	0.820	15.24	0.905	8.919	0.843
w/o UCL framework	31.490	0.870	6.694	0.926	6.676	0.894
w/o full-resolution loss	33.165	0.889	6.667	0.950	5.681	0.918
w/o VO process	33.663	0.909	6.003	0.954	5.216	0.928
Proposed	**33.966**	**0.910**	**5.828**	**0.956**	**5.079**	**0.930**

Fig. 5. It can be seen that SFNLR [46] and PSDip [64] tend to over-smooth the spatial details (see the green and red boxes in Fig. 5(h) and (i)). The corresponding error images are the residual between the fused results and the ground-truth (GT).

5.2.2. *The QuickBird dataset*

The spatial resolution of the PAN and LRMS images acquired by the QuickBird (QB) sensor is 0.7 m and 2.8 m, respectively. The QB dataset includes 20 reduced-resolution test samples. The quantitative comparison among the different approaches is reported in Table 3. The proposed method achieves the best performance. The visual performance is shown in Fig. 6. The edge details are relevant for the fused images. GDD

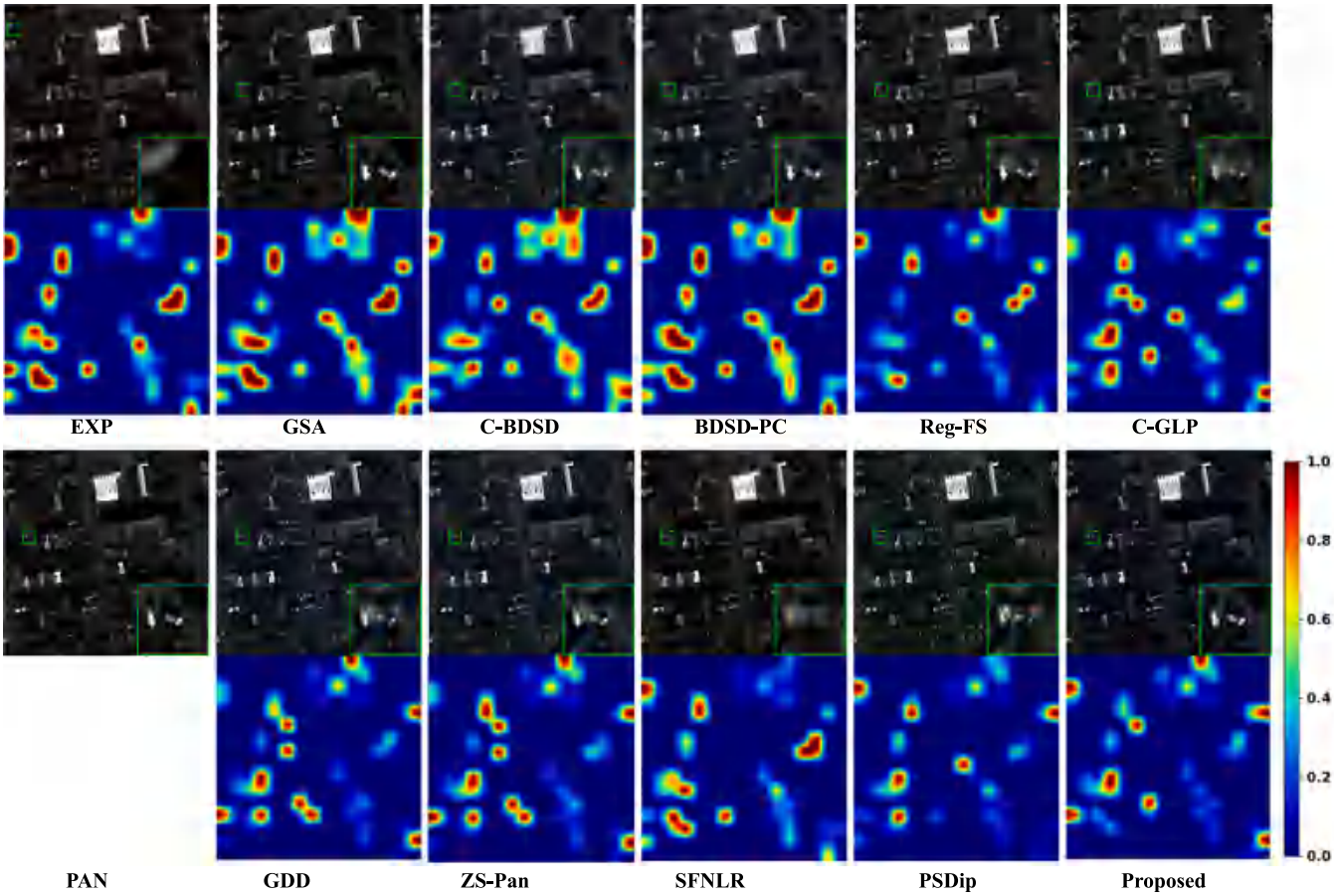

Fig. 9. Top: true color images consisting of the 3rd (R), 2nd (G), and 1st (B) bands of the fused QuickBird (QB) dataset (full-resolution), obtained by EXP, GSA, C-BDSD, BDSD-PC, Reg-FS, C-GLP, GDD, ZS-Pan, SFNLR, PSDip, proposed, and GT. Bottom: D_λ maps of the corresponding methods.

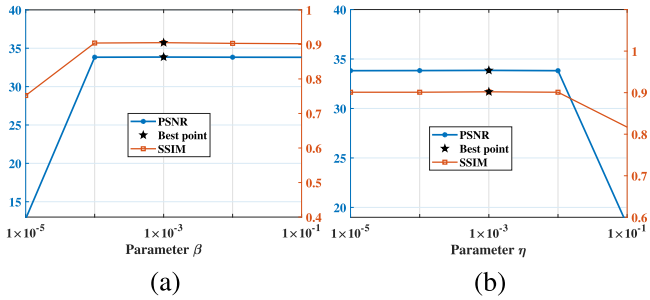

Fig. 10. Parameter analysis (using the PSNR and SSIM metrics) for β and η.

[61] and ZS-Pan [62] produce some spatial distortions on some edge areas. PSDip [64] cannot preserve details of the buildings. The proposed method is the closest to the ground-truth image.

5.2.3. *The WorldView-3 dataset*

The spatial resolution of the PAN and LRMS images acquired by the WorldView-3 (WV-3) sensor is 0.31 m and 1.24 m, respectively. On one hand, in Table 4, we report the quantitative results on 20 reduced-resolution test samples. Table 4 demonstrates that the proposed method produces the best results. On other hand, we assess the visual quality in Fig. 7. It is clear that the state-of-the-art VO method PSDip [64] produces spectral distortion. Other methods, e.g., Reg-FS [71] and C-GLP

Table 7
Quantitative results for 20 test images acquired by the WorldView-3 (WV-3) sensor and 20 test images acquired by the WorldView-2 (WV-2) sensor. All DL methods are trained with data acquired by the WV-3 sensor. (Bold: best; Underline: second best).

Dataset	Method	PSNR (± std)	SSIM (± std)	SAM (± std)	SCC (± std)	ERGAS (± std)	Q8 (± std)
WV-3	FusionNet [80]	38.09 ± 2.41	0.967 ± 0.01	3.29 ± 0.64	0.978 ± 0.02	2.433 ± 0.60	0.905 ± 0.09
	GTP-PNet [34]	37.44 ± 2.68	0.964 ± 0.01	3.50 ± 0.67	0.976 ± 0.02	2.622 ± 0.61	0.903 ± 0.09
	LAGConv [81]	**38.56 ± 2.60**	**0.971 ± 0.01**	**3.08 ± 0.49**	**0.981 ± 0.02**	**2.283 ± 0.58**	**0.911 ± 0.09**
	SFNLR [46]	33.70 ± 2.82	0.901 ± 0.03	4.74 ± 1.48	0.942 ± 0.03	4.585 ± 1.45	0.827 ± 0.11
	Proposed	35.34 ± 2.21	0.934 ± 0.02	4.15 ± 1.17	0.961 ± 0.03	3.466 ± 1.15	0.868 ± 0.10
WV-2	FusionNet [80]	28.54 ± 2.04	0.778 ± 0.04	6.41 ± 0.77	0.907 ± 0.02	5.121 ± 0.52	0.789 ± 0.08
	GTP-PNet [34]	29.37 ± 1.82	0.815 ± 0.03	6.29 ± 0.97	0.933 ± 0.01	4.668 ± 0.43	0.813 ± 0.03
	LAGConv [81]	27.93 ± 1.98	0.812 ± 0.04	6.92 ± 0.40	0.930 ± 0.02	5.292 ± 0.30	0.812 ± 0.09
	SFNLR [46]	30.51 ± 1.77	0.838 ± 0.03	5.59 ± 0.75	0.939 ± 0.02	4.176 ± 0.69	0.850 ± 0.09
	Proposed	**31.95 ± 1.80**	**0.883 ± 0.02**	**5.03 ± 0.60**	**0.957 ± 0.01**	**3.525 ± 0.51**	**0.879 ± 0.09**
	Ideal value	**+∞**	**1**	**0**	**1**	**0**	**1**

[72], cannot preserve well the complex spatial details. Our approach can retain more spatial information compared with the other methods.

5.3. Full-resolution assessment

5.3.1. The GaoFen-2 full dataset

We test the compared methods on the GaoFen-2 full dataset (size of the PAN image: 512×512). Table 5 shows that the proposed method achieves the best HQNR value. Besides, Fig. 8 provides the visual comparison and the corresponding D_λ map. As shown in Fig. 8, the proposed method can preserve more spatial details compared with the other methods. Besides, Table 5 further shows the effectiveness of the proposed method.

5.3.2. The QuickBird full dataset

In this subsection, we focus on the full-resolution assessment using data from the QB sensor. The full-resolution QB dataset consists of 20 image pairs, with no GT image available. As shown in Table 5, the proposed method achieves the best results in terms of HQNR, indicating superior overall accuracy. Fig. 9 presents a visual comparison among the different methods. Since no GT is available, we compare the results with the PAN image to assess the quality of the spatial features of the fused products. As depicted in Fig. 9, the proposed method effectively preserves spatial details.

5.4. Discussion

5.4.1. Analysis of parameters

In this subsection, we analyze the sensitivity of the hyperparameters in the proposed algorithm. The algorithm contains two hyperparameters, i.e., β and η. To assess the influence of these parameters, we plot the curve of each parameter while fixing the other. For convenience, we use two quality metrics (i.e., PSNR and SSIM). As shown in Fig. 10, the influence of β is relatively limited in the range $[10^{-3}, 10^{-1}]$, and similarly, the effect of η is limited in the range $[10^{-5}, 10^{-2}]$. These results indicate that the model's performance is robust to variations of these parameters, as it performs well across a wide range of values. To strike a balance between the different terms in the proposed model, we set $\beta = 0.001$ and $\eta = 0.001$ for the experimental analysis.

5.4.2. Ablation study

In this subsection, we analyze the impact of the proposed UCL framework, the full-resolution loss, and the variational iterative process. To evaluate their contributions, we replace the UCL framework with the coefficient estimation method from SFNLR [46]. As shown in Table 6, compared to the previous linear estimation method, the UCL framework provides more accurate coefficient estimation, leading to improved performance. Additionally, to verify the effectiveness of the proposed full-resolution loss, we retain the structure of the convolutional neural network, f_Θ, and substitute the proposed loss with the spatial loss in GDD [61]. The proposed loss can constrain the convolutional neural network, f_Θ, more effectively. Besides, the VO process further enhances the results with its iterative optimization. As illustrated in Fig. 11, the VO process consistently improves the performance throughout the iterative steps. Therefore, the proposed method, which combines both the UCL framework and the VO process, achieves the best overall performance.

5.4.3. Comparison with advanced supervised DL methods

To well describe the characteristics of the different kinds of methods, we also compare the proposed method with advanced supervised DL methods. These techniques are trained on the WV-3 dataset and tested on the WV-2 and the WV-3 datasets, respectively. As shown in Table 7, these supervised DL techniques show a relevant fusion ability. However, the excellent performance is achieved by training with 5.76GB WV-3 data. Once the test data is from other satellites, e.g., WV-2, or, anyway,

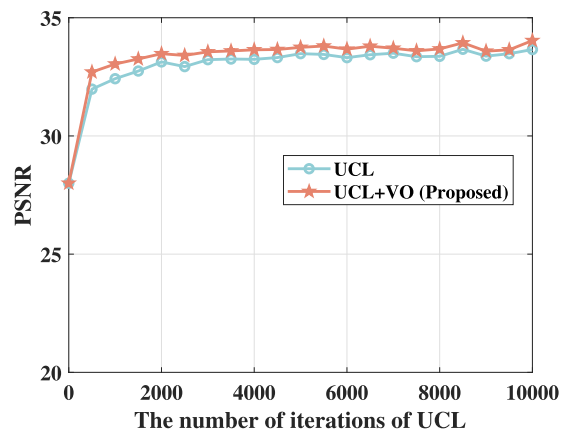

Fig. 11. The performance of the UCL framework and the proposed variational method at different iterations.

Table 8
The performance of the MRA scheme using the UCL framework and the traditional MRA approaches on a QB reduced-resolution image. (Bold: best; Underline: second best).

Method	PSNR	SSIM	SAM	SCC	ERGAS	Q4
Reg-FS [71]	30.690	0.850	7.054	0.913	7.303	0.868
C-GLP [72]	30.260	0.840	7.259	0.904	7.688	0.860
MRA-UCL	**31.150**	**0.859**	**7.025**	**0.920**	**6.905**	**0.889**

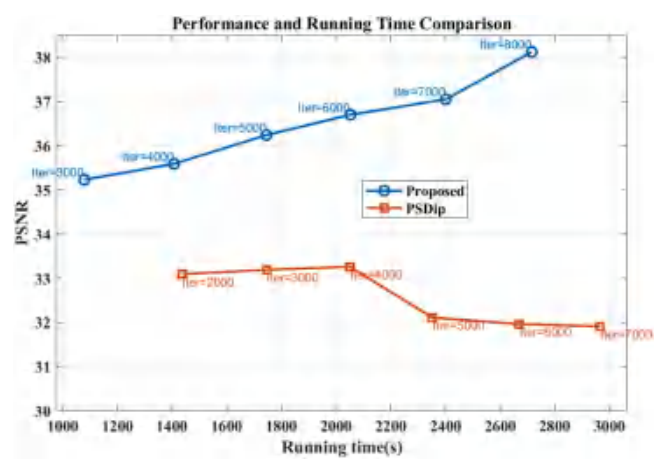

Fig. 12. Performance and time comparison on 20 GaoFen-2 reduced-resolution images. (Iter means the number of iterations).

it is inconsistent with training data, their performance severely diminishes. On the contrary, the VO approach SFNLR [46] and the proposed method only need one-pair test images and show flexibility when tested on both WV-2 and WV-3 data. More specifically, the proposed method can adaptively deal with different-source data producing outstanding results.

5.4.4. The proposed framework applied to other pansharpening schemes

The UCL framework is a general solution, which consists of two steps. First, the SkipNet (as shown in Fig. 3) takes the duplicated input and estimates the coefficients by minimizing Eq. (12). Then, the pansharpening method is applied using the estimated coefficients to generate the outcome. To evaluate the extensibility of the proposed UCL framework, we apply it to the MRA-based pansharpening scheme. As introduced in Section 1, MRA methods have shown excellent performance for spectral information extraction. These methods can be mathematically represented

Table 9
Quantitative results for 20 test images acquired by the WorldView-3 (WV-3) sensor and 20 test images acquired by the WorldView-2 (WV-2) sensor. Deep neural networks involved in LNM-PS [45] and IASF [82] are both trained with data acquired by the WV-3 sensor. (Bold: best; Underline: second best).

Dataset	Method	PSNR(± std)	SSIM(± std)	SAM(± std)	SCC(± std)	ERGAS(± std)	Q8 (± std)
WV-3	LNM-PS [45]	36.42 ± 2.50	0.953 ± 0.01	3.89 ± 0.81	0.968 ± 0.03	2.974 ± 0.78	0.889 ± 0.09
	IASF_LAGConv [82]	**38.58 ± 2.60**	**0.971 ± 0.01**	**3.08 ± 0.49**	**0.981 ± 0.02**	**2.276 ± 0.57**	**0.911 ± 0.09**
	Proposed	35.34 ± 2.21	0.934 ± 0.02	4.15 ± 1.17	0.961 ± 0.03	3.466 ± 1.15	0.868 ± 0.10
WV-2	LNM-PS [45]	29.66 ± 2.00	0.816 ± 0.04	5.67 ± 0.65	0.928 ± 0.02	4.539 ± 0.48	0.833 ± 0.09
	IASF_LAGConv [82]	29.07 ± 1.99	0.820 ± 0.04	6.33 ± 0.41	0.930 ± 0.02	4.727 ± 0.37	0.825 ± 0.09
	Proposed	**31.95 ± 1.80**	**0.883 ± 0.02**	**5.03 ± 0.60**	**0.957 ± 0.01**	**3.525 ± 0.51**	**0.879 ± 0.09**
	Ideal value	**+∞**	**1**	**0**	**1**	**0**	**1**

as:

$$\mathcal{X} = \widetilde{\mathcal{Y}} + \mathcal{G}_{in} \odot (\overline{\mathcal{P}} - \overline{\mathcal{P}}_L), \tag{23}$$

where $\mathcal{G}_{in} \in \mathbb{R}^{H\times W\times S}$ is the injection coefficient, $\overline{\mathcal{P}}$ denotes the histogram-matched PAN image, and $\overline{\mathcal{P}}_L$ is the low-pass version of $\overline{\mathcal{P}}$. Many MRA methods have been proposed by designing different injection coefficients, e.g., Reg-FS [71] and C-GLP [72]. We estimate the injection coefficient $\mathcal{G}_{in}$ by the proposed UCL framework. The corresponding full-resolution loss is as follows:

$$Loss_{MRA-UCL} = \left\|\nabla f_{\Theta}(\mathcal{P}) - \nabla(\mathcal{H}_1 \oslash \mathcal{H}_2)\right\|_F^2, \tag{24}$$

where $\mathcal{H}_1 = (\Gamma_\theta(\mathcal{P}) - \widetilde{\mathcal{Y}})$ and $\mathcal{H}_2 = (\overline{\mathcal{P}} - \overline{\mathcal{P}}_L)$. As shown in Table 8, our UCL framework can accurately estimate the coefficient and shows its excellent flexibility in solving pansharpening problems using different schemes.

5.4.5. Time analysis

To analyze the time consumption, we compare the proposed method to the state-of-the-art method PSDip [64]. As shown in Fig. 12, the proposed method is more efficient. With the increase of the number of iterations and time consumption, the performance of our method also increases accordingly. Besides, the performance of the proposed method with a reduced running time is more powerful than the best result of PSDip [64]. Meanwhile, Fig. 12 also demonstrates that the proposed method can significantly improve running time with only a marginal drop in performance by lowering the number of iterations, highlighting its practical flexibility.

5.4.6. Comparison with spatial-fidelity-based VO approaches

To highlight the advantages of the proposed method over other spatial-fidelity-based VO approaches, namely LNM-PS [45] and IASF_LAGConv [82], we conduct experiments on both the WV-2 and WV-3 datasets. It is worth noting that IASF_LAGConv [82] relies on a pre-trained deep neural network (i.e., LAGConv [81]), while LNM-PS [45] requires additional data to train a nonlinear mapping-based spatial fidelity model. In the experiment, both the deep neural network and the nonlinear mapping are trained exclusively on the WV-3 training data. As reported in Table 9, these spatial-fidelity-based VO methods demonstrate strong performance when the test data are consistent with the training distribution. However, their performance deteriorates significantly on the WV-2 dataset due to the gap between the test and training data. In contrast, the proposed method exhibits stable and robust performance across both datasets, highlighting its generalization ability and independence from additional training data.

6. Conclusion

In this paper, we presented a novel variational method via a general UCL framework. More specifically, we first estimate the coefficient needed in the variational model using the UCL framework. Afterward, an iteration update algorithm has been designed to further optimize the final result. The experiments on reduced- and full-resolution data verified the high performance and flexibility of the proposed method. Besides, we analyzed the hyperparameters, we provided an ablation study, we conducted the comparisons with advanced supervised DL methods and spatial-fidelity-based VO approaches, and we extended the proposed UCL framework to another pansharpening scheme. Future developments go toward the use of the UCL framework exploring other image processing tasks, e.g., image denoising and image restoration.

CRediT authorship contribution statement

Jin-Liang Xiao: Writing - original draft; **Ting-Zhu Huang:** Methodology, Funding acquisition; **Liang-Jian Deng:** Writing - review & editing; **Huidong Jiang:** Visualization; **Qibin Zhao:** Writing - review & editing; **Gemine Vivone:** Writing - review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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